

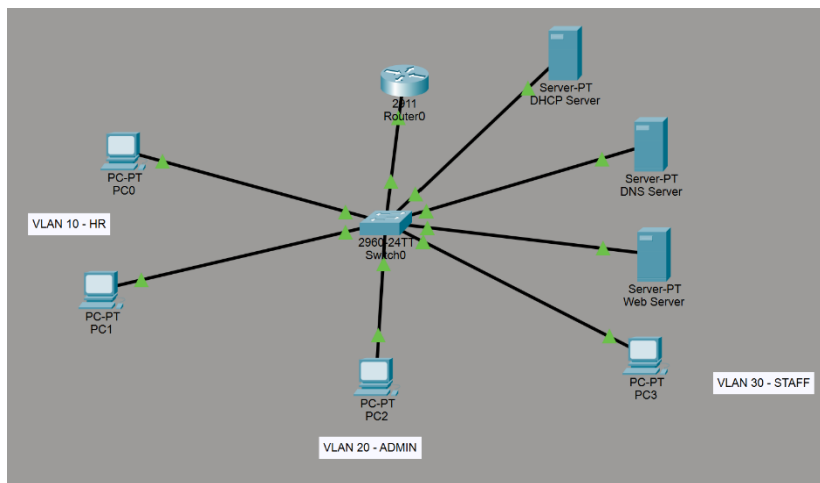
Enterprise Network Design Using Cisco Packet Tracer

1. Network Topology

Overview

The network was designed to simulate a small enterprise environment using Cisco Packet Tracer. The network topology consists of a router connected to a switch with multiple PCs connected to the switch. Three departments like HR, Admin and Staff were created using VLANs to logically separate network traffic while sharing the same physical infrastructure.

The router provides Layer 3 connectivity between the VLANs using the Router on a Stick approach. Additional network services including DHCP, DNS and web server were implemented to automate IP address allocation and provide hostname resolution.



2. VLAN Implementation

Overview

Three Virtual Local Area Networks (VLANs) were created to separate the departments within the organisation. Each VLAN represents a different broadcast domain and has its own IP network.

VLAN	Department	Network
VLAN 10	HR	192.168.10.0/24
VLAN 20	Admin	192.168.20.0/24
VLAN 30	Staff	192.168.30.0/24

Switch ports were assigned to the appropriate VLANs, ensuring that devices connected to those ports become members of the correct department.

The show vlan brief command was used to display the configured VLANs and verify that the correct switch ports were assigned to each VLAN.

3. Inter-VLAN Routing

Overview

Inter-VLAN routing was implemented using the Router on a Stick method. A trunk connection was configured between the switch and the router. Router sub interfaces were created for each VLAN and configured with encapsulation. Each sub interface was assigned an IP address that acts as the default gateway for devices within its VLAN.

This configuration allows devices located in different VLANs to communicate while maintaining separation between departments.

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
!
```

Output of the show vlan brief command showing VLAN 10, VLAN 20, VLAN 30.

```
Switch>show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Gig0/1    on        802.1q         trunking      1

Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20,30
```

Output of the show interfaces trunk command confirming the trunk link between the switch and router.

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Connectivity tests were performed from the HR (VLAN 10) network by pinging another device within the same VLAN and devices located in the Admin (VLAN 20) and Staff (VLAN 30) networks.

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time=7ms TTL=127

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 1ms

C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Connectivity tests were performed from the Admin (VLAN 20) network by pinging devices within the HR (VLAN 10) and Staff (VLAN 30) networks.

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Connectivity tests were performed from the Staff (VLAN 30) network by pinging devices within the HR (VLAN 10) and Admin (VLAN 20) networks.

4. DHCP Implementation

Dynamic Host Configuration Protocol (DHCP) was implemented to automatically assign IP addresses and network settings to client devices within each VLAN. Separate DHCP pools were created for VLAN 10 (HR), VLAN 20 (Admin), and VLAN 30 (Staff). Each pool was configured with the appropriate network settings, and DHCP relay was configured on the router to forward client requests to the DHCP server.

Servers were assigned static IP addresses because network services such as DHCP, DNS, and the web server require a permanent address so that client devices can always locate them. Client PCs were configured to use DHCP because it automatically provides network settings and reduces the need for manual configuration.

Step 1 – Configure the DHCP Server

The DHCP server was assigned a static IP address so that it could provide IP address allocation services to all VLANs.



Step 2 – Create DHCP Pools

Three DHCP pools were created for VLAN 10 (HR), VLAN 20 (Admin), and VLAN 30 (Staff). Each pool was configured with the appropriate default gateway, subnet mask, DNS server, and starting IP address to provide automatic IP address allocation for devices within each VLAN.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
HR	192.168.10.1	192.168.30.6	192.168.10.2	255.255.255.0	254	0.0.0.0	0.0.0.0
Staff	192.168.30.1	192.168.30.6	192.168.30.2	255.255.255.0	254	0.0.0.0	0.0.0.0
Admin	192.168.20.1	192.168.30.6	192.168.20.2	255.255.255.0	254	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.30.0	255.255.255.0	512	0.0.0.0	0.0.0.0

Step 3 – Configure DHCP Relay

The router was configured with the ip helper-address command on each VLAN sub interface to forward DHCP requests from clients to the DHCP server.

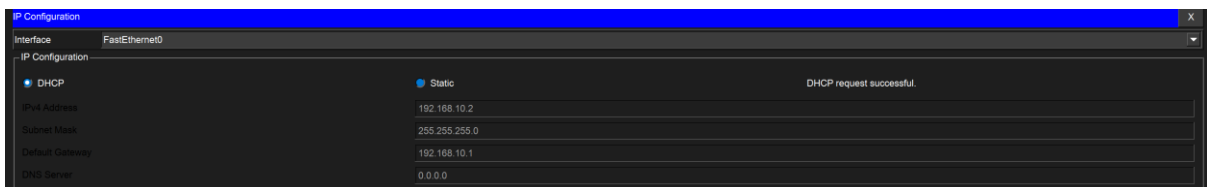
```

interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.30.5
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 ip helper-address 192.168.30.5
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
.

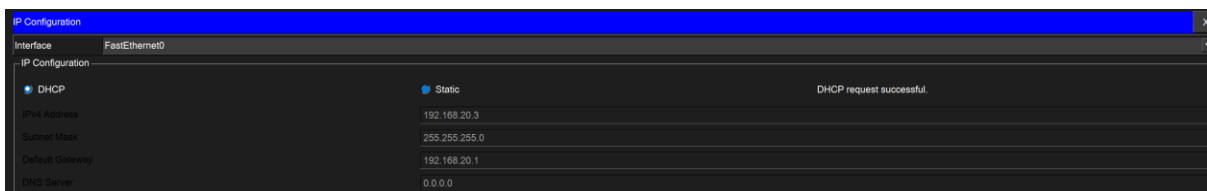
```

Step 4 – Configure Client Devices

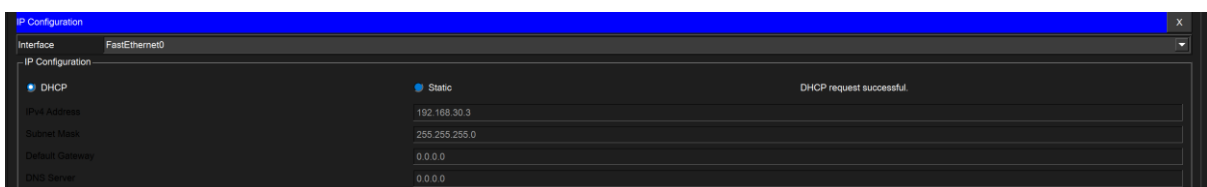
Each client device was configured to obtain its IP address automatically by changing the IP configuration from Static to DHCP. This allowed the devices to request their network configuration from the DHCP server.



The images show the DHCP request successful for the HR Network



This image shows the DHCP request successful for the Admin Network



This image shows the DHCP request successful for the Staff Network

5. DNS and Web Server Implementation

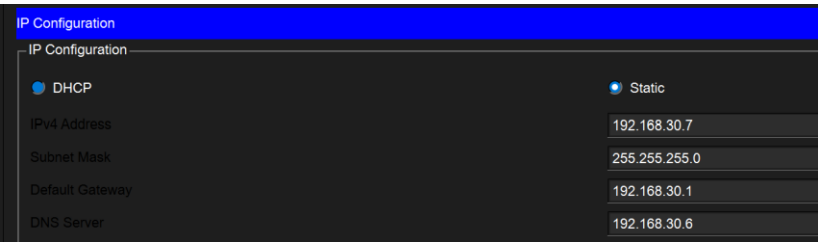
Overview

The Domain Name System (DNS) and a web server were used to demonstrate hostname resolution across the network. A dedicated web server was configured to host an internal

website, while a dedicated DNS server was used to translate the hostname companylocal.com into the IP address of the web server. This allows users to access the website using its hostname instead of the IP address.

Step 1 – Configure the Web Server

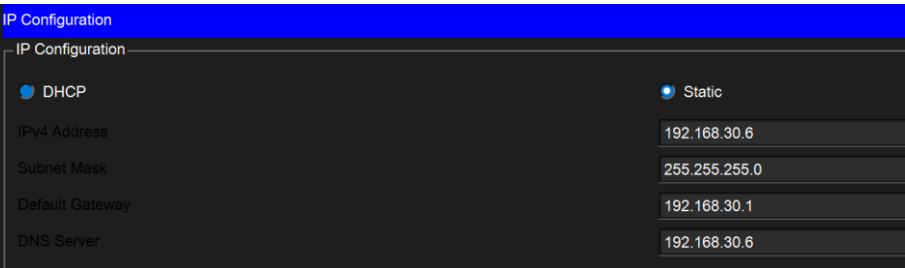
A web server was added to the network and configured with a static IP address. The server was assigned the correct IP address, subnet mask, default gateway, and DNS server to allow communication with other devices. The HTTP service was enabled to host an internal company website.



This image shows the web server IP configuration.

Step 2 – Configure the DNS Server

A dedicated DNS server was added to the network and assigned a static IP address. The DNS service was enabled to provide hostname resolution.



This image shows the DNS server IP configuration.

Step 3 – Create the DNS Record

A DNS record was created to map the hostname companylocal.com to the IP address of the web server.

No.	Name	Type	Detail
0	companylocal.com	A Record	192.168.30.7

Step 4 – Configure Client DNS Settings

Client devices automatically received the DNS server address through the DHCP configuration. The assigned DNS settings were verified using the client network configuration, confirming that devices were correctly configured to use the DNS server for hostname resolution.

Step 5 – Verify DNS and Web Server Operation

The DNS configuration was verified by successfully pinging companylocal.com from a client device. The web server was then verified by opening a web browser and navigating to <http://companylocal.com>. The successful loading of the webpage confirmed that the hostname was correctly resolved and that the web server was operating correctly.

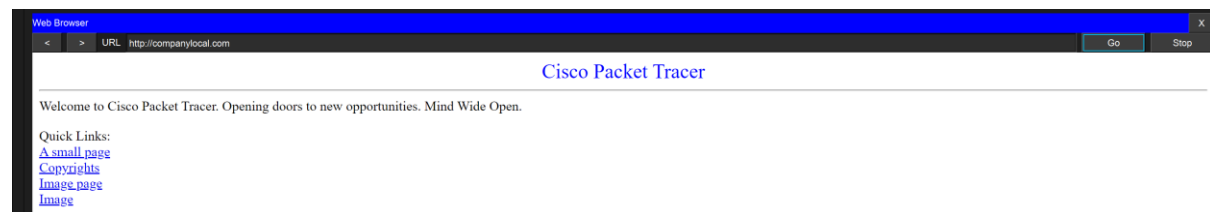
```
C:\>ping companylocal.com

Pinging 192.168.30.7 with 32 bytes of data:

Reply from 192.168.30.7: bytes=32 time=1ms TTL=127
Reply from 192.168.30.7: bytes=32 time<1ms TTL=127
Reply from 192.168.30.7: bytes=32 time<1ms TTL=127
Reply from 192.168.30.7: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

This shows the successful ping of companylocal.com



This screenshot shows the successful loading of the internal website at <http://companylocal.com> from a client PC.

The enterprise network was successfully implemented using VLANs, inter-VLAN routing, DHCP, DNS and web services. Testing confirmed that devices could communicate between networks, receive automatic IP addresses, and access services using hostnames.